

ContextFusion — Multi-Sensor Counter-UAV System Architecture

Sensing

Detection

Fusion & Tracking

Reasoning

Output

Camera

YOLOv11m · 25 Hz

Object
Detector

BBox · conf · class

MAP Fusion

IoU + conf + SNR

XAI Threat
Scoring

r·v·c factors
SHAP attribution

Operator
Display

SHAP bars · track list
threat ranking

AERIS-10 Radar

FMCW X-band
f_x=10 GHz · 10 Hz
CA-CFAR detection

Radar
Processor

Range · bearing
SNR · Doppler
Swerling RCS

Kalman
Tracker

CV model · Δt=0.5 s
4-state [x,y,ẋ,ẏ]
gating θ≤15°

WTA
Optimizer

Hungarian alg.
τ_i × P_{k^{ij}} cost
battery constraints

Gimbal
Control

MAVLink
bearing → motor
18 FPS end-to-end

Primary data flow Auxiliary measurement path